# Reconstructing $\gamma \rightarrow e^+e^-$ with the sPHENIX detector at RHIC

*Valerie Wolfe, Lehigh University, for the sPHENIX Collaboration*


## Abstract

A photon encountering high density matter will probabilistically undergo pair production, resulting in a lepton-antilepton pair that can be detected to reconstruct the incident photon. This analysis uses proton-proton data collected by the sPHENIX detector to reconstruct conversion photons via the dielectron channel. Tracks are fitted to data collected by the sPHENIX Monolithic Active Pixel Vertex detector (MVTX), Silicon Strip Intermediate Tracker (INTT), and Time Projection Chamber (TPC). Electron and positron candidates are identified using an invariant mass analysis. Tracks identified as daughter candidates are matched to energy clusters in the Electromagnetic Calorimeter (EMCal), and momenta are extracted from the track curvature. The purity of the daughter candidates is verified using the energy to momentum ratio, E/p. This analysis sets the groundwork for searches for higher mass resonances reconstructed using the dielectron channel, enabling the reconstruction of charmonia and bottomonia, a vital component of the sPHENIX heavy-flavor program.

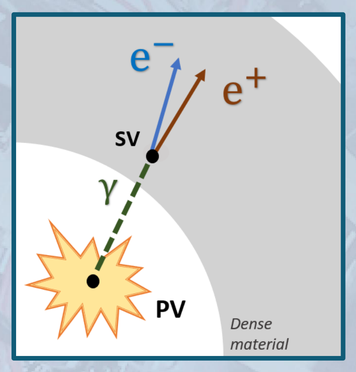




Figure 1: An illustration showing the initial collision at the PV and the conversion at the SV.

## Physics Background

- Protons collide at the primary vertex (PV), producing new particles (fig. 1)
- Outgoing photons encounter dense matter and convert into two leptons via pair production at the secondary vertex (SV)
- Photon is massless → Angle between daughters is effectively zero

## sPHENIX Data Collection

- sPHENIX collected proton+proton (pp) data at $\sqrt{s} = 200$ GeV during Run 3 (Dec. 2025 – Jan. 2026)
- Position data were collected by the tracking subsystems, MVTX, INTT, and TPC (fig. 2)
- Energy measured by EMCal
  - $e^{+/-}$ create electromagnetic showers (fig. 2)
  - $e^{+/-}$ deposit (nearly) all their energy → $E/p \approx 1$

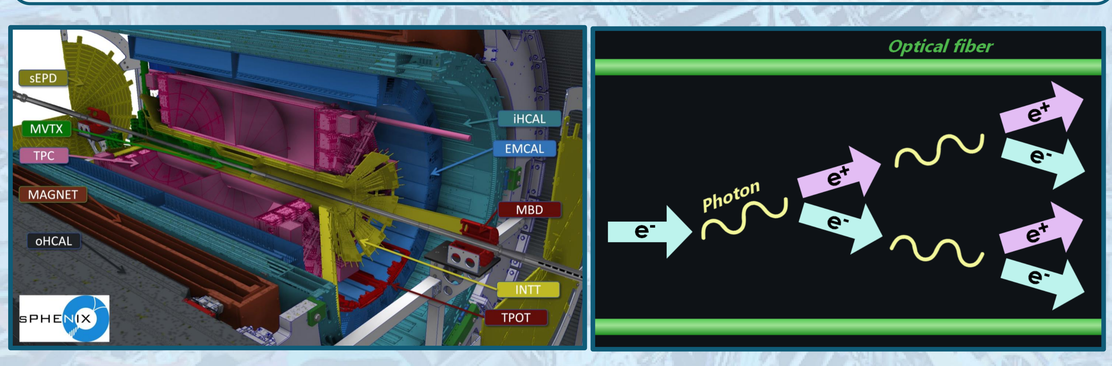



Figure 2: A schematic of the sPHENIX detector (left) and an illustration of electromagnetic showers in the EMCal (right).

| Effect | Cut values |
|---|---|
| Small invariant mass | $M_{\text{inv}} < 0.05$ GeV/c$^2$ |
| Opening angle between tracks near zero | $\lvert\varphi_{e-} - \varphi_{e+}\rvert < 0.01$ rad<br>$\lvert\eta_{e-} - \eta_{e+}\rvert < 0.02$ |
| Conversion radius must be in material | $0.5 < r_{SV} < 30$ cm |

Table 1: A table summarizing cuts made to reject background.

## Procedure

- Position data fitted with tracks
- Momenta are extracted from tracks curvature
- Energy deposits in EMCal grouped into clusters
- KFParticle identifies candidate track pairs with an invariant mass analysis
- Tracks are matched to EMCal clusters
- Additional cuts are applied to reject background (table 1)

## Track-EMCal Matching

1. Track $i$ "projected" to a state in the EMCal at $r_{state} = 102.9$ cm
2. The state position is compared to each cluster center
3. The cluster which minimizes distance in $\eta - \varphi$ space is selected as the match for track $i$
4. Daughter pairs missing one or both cluster matches are removed from dataset

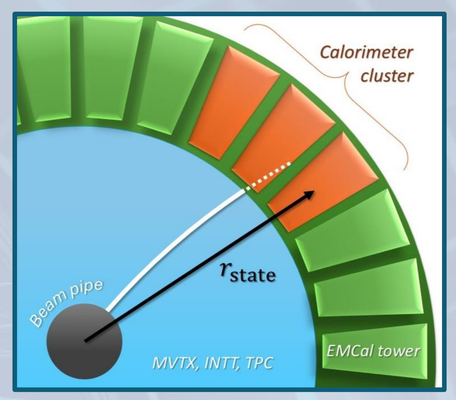



Figure 3: An illustration showing track projection into the EMCal. The solid white line indicates a fitted track; the dashed white line indicates track projection.

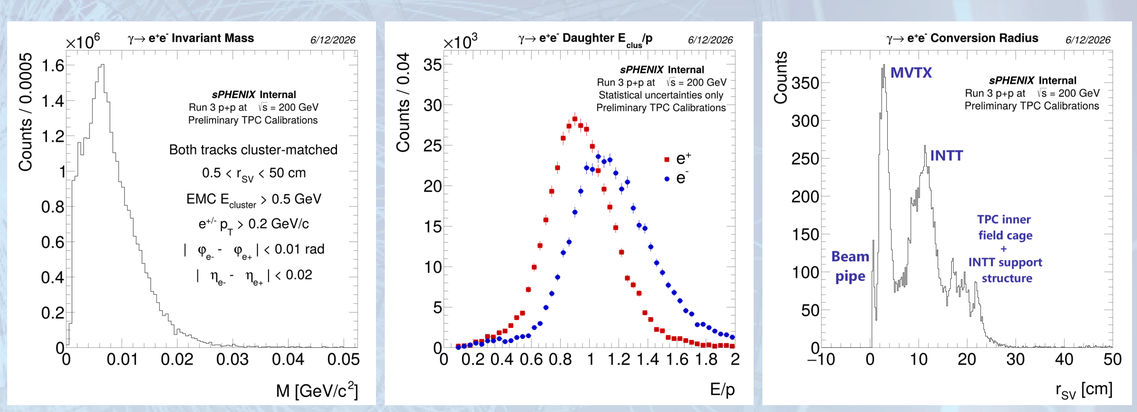



Figure 4: Distributions of $M_\gamma$ (left), daughter $E/p$ (middle), and $r_{SV}$ (right). Calibrations applied to the data are preliminary; in particular, the TPC calibrations do not yet account for space charge distortions.

## Results and Conclusions

- Peak of $M_{\text{inv}} \sim 0$ and $r_{SV}$ peaks line up with material of detector

*Selections are effectively isolating photon conversions.*

- Daughter $E/p$ distributions are gaussian with peaks at $\sim 1$

*Matching tracks to EMCal clusters is successfully demonstrated for Run 3 pp data, enabling quarkonia reconstruction in this new dataset.*

- Positron/electron $E/p$ mean shifted slightly below/above one
- Note: previous results from Run 2 pp show smaller separation between peaks

*The upgrade to the TPC after Run 2 added new complexity to the calibrations. These calibrations are in preliminary stages, so the track-cluster matching performance is expected to improve when final stage calibrations are available.*


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